

Copula Entropy

Theory and Applications

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Causal Discovery¹

¹Jian Ma. "Estimating Transfer Entropy via Copula Entropy". In: *arXiv preprint arXiv:1910.04375* (2019).

Copula Entropy: Causal Discovery

- Problem
 - To infer causality from time series data by *estimating Transfer Entropy*
- History & Significance
 - Causality is one of the oldest topics in philosophy
 - Causal discovery is a fundamental problem of all sciences
- Correlation vs Causality
 - Correlation does not mean causation
 - Correlation is symmetric while causality is asymmetric
 - Correlation is only helpful for prediction while causality implies temporal relation, intervention and control

Copula Entropy: Causal Discovery

- **Wiener's Principle²**

Cause should improve the prediction of effect.

²[Norbert Wiener](#). "The theory of prediction. Modern mathematics for engineers". In: *New York* 165.6 (1956).

Copula Entropy: Causal Discovery

- **Granger Causality**³

improvement measured by the variance of prediction error

Definition (Granger Causality)

Given time series X_t, Y_t , we say X cause Y in Granger's sense if

$$\delta^2(Y_{t+1}|Y_t, X_t) < \delta^2(Y_{t+1}|Y_t), \quad (1)$$

where δ^2 is variance of prediction error.

³Clive WJ Granger. "Investigating causal relations by econometric models and cross-spectral methods". In: *Econometrica: Journal of the Econometric Society* (1969), pp. 424–438, Clive WJ Granger. "Testing for causality: a personal viewpoint". In: *Journal of Economic Dynamics and control* 2 (1980), pp. 329–352.

Copula Entropy: Causal Discovery

Transfer Entropy⁴

- Definition

improvement on the uncertainty of prediction measured by Shannon entropy

$$TE = \sum p(Y_{t+1}, Y^t, X_t) \log \frac{p(Y_{t+1} | Y^t, X_t)}{p(Y_{t+1} | Y^t)} \quad (2)$$

$$= H(Y_{t+1} | Y^t) - H(Y_{t+1} | Y^t, X_t) \quad (3)$$

⁴Thomas Schreiber. "Measuring information transfer". In: *Physical Review Letters* 85.2 (2000), pp. 461–464.

Copula Entropy: Causal Discovery

Transfer Entropy

- Essentially conditional mutual information

$$TE = I(Y_{t+1}; X_t | Y^t) \quad (4)$$

Copula Entropy: Causal Discovery

Transfer Entropy⁵

- Entropy based estimation

$$TE = H(Y_{t+1}, Y_t) + H(Y_t, X_t) - H(Y_{t+1}, Y_t, X_t) - H(Y_t) \quad (5)$$

- Issue
difficult to estimate, some think impossible without model assumptions

⁵Terry Bossomaier et al. *An Introduction to Transfer Entropy: Information Flow in Complex Systems*. Springer Cham, 2016.

Copula Entropy: Causal Discovery

TE via CE⁶

Proposition

$$T_{x \rightarrow y} = -H_c(Y_{t+1}, Y^t, X_t) + H_c(Y_{t+1}, Y^t) + H_c(Y^t, X_t) - H_c(Y^t) \quad (6)$$

Proof.

$$TE = \sum p(Y_{t+1}, Y^t, X_t) \log \frac{p(Y_{t+1} | Y^t, X_t)}{p(Y_{t+1} | Y^t)} \quad (7)$$

$$= \sum p(Y_{t+1}, Y^t, X_t) \log \frac{p(Y_{t+1}, Y^t, X_t) p(Y^t)}{p(Y_{t+1}, Y^t) p(Y^t, X_t)} \quad (8)$$

$$= I(Y_{t+1}; Y^t; X_t) - I(Y_{t+1}; Y^t) - I(Y^t; X_t) \quad (9)$$

$$= -H_c(Y_{t+1}; Y^t; X_t) + H_c(Y_{t+1}; Y^t) + H_c(Y^t; X_t) \quad (10)$$

$$= -H_c(Y_{t+1}, Y^t, X_t) + H_c(Y_{t+1}, Y^t) + H_c(Y^t, X_t) - H_c(Y^t) \quad (11)$$



⁶Jian Ma. "Estimating Transfer Entropy via Copula Entropy". In: *arXiv preprint arXiv:1910.04375* (2019).

Copula Entropy: Causal Discovery

TE via CE⁷

Proposition

Transfer Entropy can be represented with only Copula Entropy.

$$T_{x \rightarrow y} = -H_c(Y_{t+1}, Y^t, X_t) + H_c(Y_{t+1}, Y^t) + H_c(Y^t, X_t) - H_c(Y^t) \quad (12)$$

- Non-parametric Estimator of TE
 - ① estimating three or four CE terms in (12);
 - ② calculating TE for these estimated CEs.
- inheriting all the merits of the non-parametric CE estimator

⁷ Jian Ma. "Estimating Transfer Entropy via Copula Entropy". In: *arXiv preprint arXiv:1910.04375* (2019).

Copula Entropy: Causal Discovery

Experiments on the UCI Beijing PM2.5 data

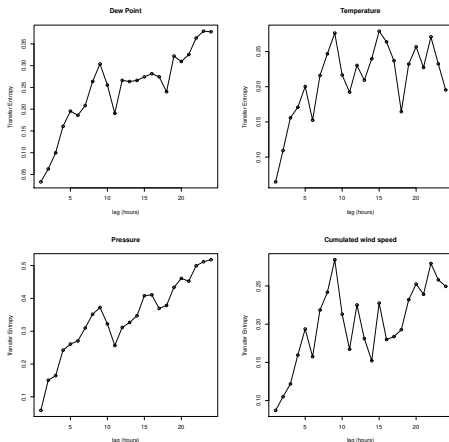
- Overview of the data
 - Time

hourly data from 2010-01-01 to 2014-12-31, which results in 43824 samples with missing values.
 - Observations
 - PM2.5 data of US Embassy in Beijing
 - Meteorological data from Beijing Capital International Airport
 - Meteorological factors

dew point, temperature, pressure, cumulated wind speed, combined wind direction, cumulated hours of snow, cumulated hours of rain.
- Experimental data
 - the first four factors used in the experiments;
 - 1000 samples without missing values (2010-04-02~2010-05-14).

Copula Entropy: Causal Discovery

Results: Effects of meteorological factors on PM2.5



• Two phrases

- Sharp increase phrase: the first 9 hours time lag, and peak at about 9 hours lag;
- Flat increase phrase: TE of Dew point and pressure increase with relatively flat rate while TE of temp. and cumulated wind speed does increase any more.

• Interpretation

- The effects do not show immediately and are cumulating processes.

Copula Entropy: Application 5

Time Lag Estimation⁸

⁸ Jian Ma. "Identifying Time Lag in Dynamical Systems with Copula Entropy based Transfer Entropy". In: *arXiv preprint arXiv:2301.06037*, arXiv:2301.06037 (2023). arXiv: 2301.06037 [cs.LG].

Copula Entropy: Time Lag Estimation

- Problem
 - To identify time lag in dynamical systems with copula entropy based transfer entropy
- Significance
 - Time lag is ubiquitous in physical, social, and biological systems
 - Identifying time lag is of fundamental importance in applications of dynamical systems
- Related Methods
 - Auto-correlation
 - Time-delayed mutual information

Copula Entropy: Time Lag Estimation

- System model

$$y_t = f(x_{t-l}, \mathbf{z}_t) + \epsilon \quad (13)$$

- Our method

- ① estimating transfer entropies in time lag window from data with the CE-based estimator
- ② identifying the time lag l associated with the maximal TE value

$$\hat{l} = \arg \max_l TE_{X \rightarrow Y}(l) \quad (14)$$

Copula Entropy: Time Lag Estimation

Simulations

- ① generate trajectories from five simulated dynamical system with respect to different state or output lags
- ② identify the time lag with our method

Copula Entropy: Time Lag Estimation

Simulated systems

- ① a system driven by random input with output lag

$$\begin{aligned}x_i &= \xi_1, \\ y_{i+l} &= x_i + \xi_2;\end{aligned}\tag{15}$$

- ② a system driven by sine function with output lag

$$\begin{aligned}x_i &= \sin(2\pi i/m) + \xi_1, \\ y_{i+l} &= x_i + \xi_2;\end{aligned}\tag{16}$$

- ③ Wiener process with output lag

$$\begin{aligned}x_i &= x_{i-1} + \xi_1, \\ y_{i+l} &= x_i + \xi_2;\end{aligned}\tag{17}$$

Copula Entropy: Time Lag Estimation

Simulated systems

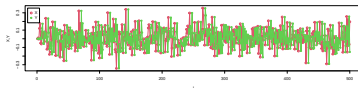
- 4 a first-order linear system with state lag

$$\begin{aligned}x_i &= \alpha x_{i-1} + \beta x_{i-l} + \xi_1, \\y_i &= x_i + \xi_2;\end{aligned}\tag{18}$$

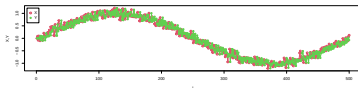
- 5 a first-order nonlinear system with state lag

$$\begin{aligned}x_i &= \alpha x_{i-1} + \beta x_{i-l} + \xi_1, \\y_i &= x_i^2 + \sin(x_i) + \xi_2.\end{aligned}\tag{19}$$

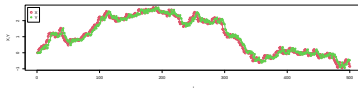
Copula Entropy: Time Lag Estimation



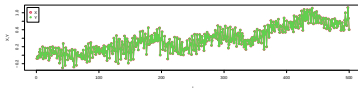
(a) random system



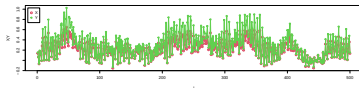
(b) sin function



(c) Wiener process



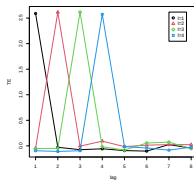
(d) linear first-order system



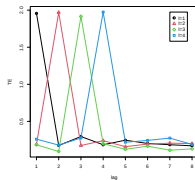
(e) nonlinear first-order system

Figure: Simulated trajectories.

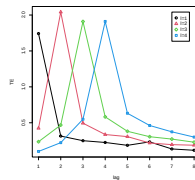
Copula Entropy: Time Lag Estimation



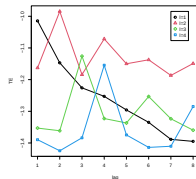
(a) random system



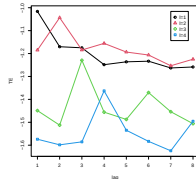
(b) sin function



(c) Wiener process



(d) linear system



(e) nonlinear system

Figure: Simulation results of the experiments on time lag estimation.

Copula Entropy: Time Lag Estimation

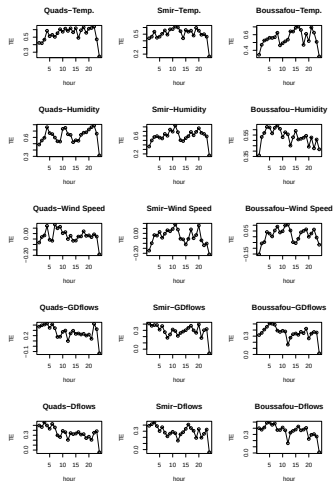
Real data: Power consumption of the Tetouan city⁹

- Data

- power consumption of 3 networks in 2017
- weather factors, including temperature, humidity, wind speed, general diffuse flows, diffuse flows

- Power consumption forecast

- To identify time lags from weather to power consumption



⁹Asuncion2007.

Copula Entropy: Application 6

System Identification¹⁰

¹⁰ Jian Ma. "Two-Sample Test with Copula Entropy". In: *arXiv preprint arXiv:2307.07247* (2023). arXiv: 2307.07247 [stat.ME].

Copula Entropy: System Identification

- Problem
 - To discover differential equation from time series data
- Significance
 - differential equations are the main mathematical tools for modeling dynamical systems
 - discovering differential equations of dynamical systems has wide applications in many scientific fields
- Related Methods
 - SINDy
 - Gaussian processes

Copula Entropy: System Identification

Sparse Identification of Nonlinear Dynamical systems (SINDy)¹¹

• Idea

use **sparse regression** to find a sparse linear combination of basis functions that best approximate the differential equations of dynamical system

$$\frac{dx}{dt} = f(x(t)). \quad (20)$$

• Method

- ① perform numerical differentiation $\frac{dx}{dt}$;
- ② specify a set of basic functions $\Theta(x)$ such that

$$\frac{dx}{dt} = \Theta(x)\xi; \quad (21)$$

③ LASSO regression

$$\hat{\xi} = \arg \min_{\xi} \|\Theta(x)\xi - \frac{dx}{dt}\| + \lambda \|\xi\|_1 \quad (22)$$

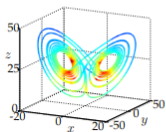
¹¹Steven L Brunton, Joshua L Proctor, and J Nathan Kutz. "Discovering governing equations from data by sparse identification of nonlinear dynamical systems". In: *Proceedings of the national academy of sciences* 113.15 (2016), pp. 3932–3937.

Copula Entropy: System Identification

Sparse Identification of Nonlinear Dynamical systems (SINDy)

I. True Lorenz System

$$\begin{aligned}\dot{x} &= \sigma(y - x) \\ \dot{y} &= x(\rho - z) - y \\ \dot{z} &= xy - \beta z.\end{aligned}$$



Data In

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} 1 & x & y & z & x^2 & xy & xz & y^2 & z^5 \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix}$$

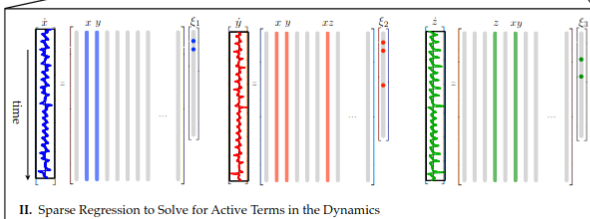
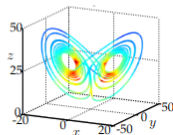
	'xi_1'	'xi_2'	'xi_3'
'1'	[0]	[0]	[0]
'x'	[-9.9996]	[27.9980]	[0]
'y'	[9.9998]	[-0.9997]	[0]
'z'	[0]	[0]	[-2.6665]
'xx'	[0]	[0]	[0]
'xy'	[0]	[0]	[1.0000]
'xz'	[0]	[-0.9999]	[0]
'yy'	[0]	[0]	[0]
'yz'	[0]	[0]	[0]
'zz'	[0]	[0]	[0]
'xxxx'	[0]	[0]	[0]
'yyyy'	[0]	[0]	[0]

Sparse Coefficients of Dynamics

Model Out

III. Identified System

$$\begin{aligned}\dot{x} &= \Theta(x^T) \xi_1 \\ \dot{y} &= \Theta(x^T) \xi_2 \\ \dot{z} &= \Theta(x^T) \xi_3\end{aligned}$$



II. Sparse Regression to Solve for Active Terms in the Dynamics

Copula Entropy: System Identification

Our idea

- consider system identification as a **variable selection** problem
- apply CE for variable selection

Copula Entropy: System Identification

Our method

- ① calculating the derivative of system variables with differential operator;

$$\left. \frac{dx}{dt} \right|_{t=t_0} \approx \frac{x_{t_1} - x_{t_0}}{t_1 - t_0}. \quad (23)$$

- ② estimating the CEs $H_c(\cdot, \cdot)$ between the calculated derivatives $\frac{dx}{dt}$ and the covariates of the system $x, y, z, \dots$;
- ③ selecting the covariates with high CE value for each derivatives $\frac{dx}{dt}$.

Copula Entropy: System Identification

Simulations

simulating time series data from Lorenz system and Rössler system

Lorenz system

$$\frac{dx}{dt} = \sigma(y - x) \quad (24)$$

$$\frac{dy}{dt} = \rho x - y - xz \quad (25)$$

$$\frac{dz}{dt} = -\beta z + xy \quad (26)$$

Rössler system

$$\frac{dx}{dt} = -(y + z) \quad (27)$$

$$\frac{dy}{dt} = x + ay \quad (28)$$

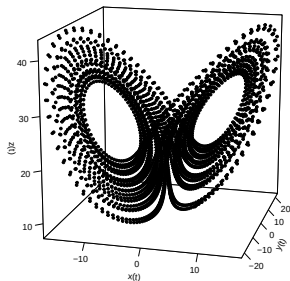
$$\frac{dz}{dt} = b + z(x - c) \quad (29)$$

Copula Entropy: System Identification

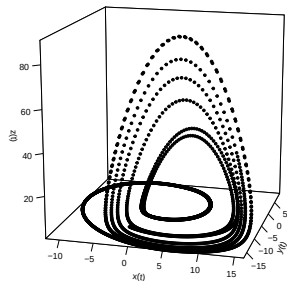
Simulations

simulating time series data from Lorenz system and Rössler system

Lorenz system



Rössler system



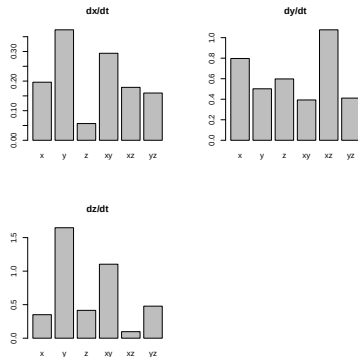
Copula Entropy: System Identification

Results - Lorenz system

$$\frac{dx}{dt} = 10(y - x)$$

$$\frac{dy}{dt} = 28x - y - xz$$

$$\frac{dz}{dt} = -\frac{8}{3}z + xy$$



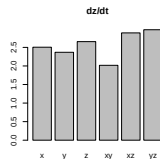
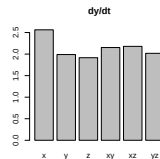
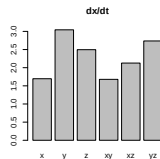
Copula Entropy: System Identification

Results - Rössler system

$$\frac{dx}{dt} = -(y + z)$$

$$\frac{dy}{dt} = x + 0.38y$$

$$\frac{dz}{dt} = 0.2 + z(x - 5.7)$$



Monograph

Jian Ma. "Copula Entropy: Theory and Applications". In: *arXiv preprint arXiv:2025.18168* (2025)

Thanks!

Q&A